

Ronaldo Franco Jaldin

Software Engineer · Machine Learning

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I'm passionate about machine learning and enjoy working through difficult problems, even when finding a solution takes weeks. I want to build software that makes a positive difference in people's everyday lives.

Education

HSE University — Moscow

Graduated July 2026

BSc, Applied Mathematics and Informatics · Started Sep 2021

Experience

Freelance Machine Learning Engineer

Apr 2025–present

- Delivered a camera-based attendance system with face detection and anti-spoofing checks.
- Developed video analytics to recognize coffee-shop products, count production, and generate timestamped reports; prepared training data with model-assisted annotation.
- Use Claude and Codex during software development.

Research Assistant — Huawei Research Institute

Mar 2024–Mar 2025

- Built reproducible test-environment pipelines across repositories and contributed Docker environments to the SWE-bench harness.
- Built a patch/file retrieval pipeline and fine-tuned Jina embeddings with QLoRA, reaching 0.62 MRR on SWE-bench Verified above the off-the-shelf baseline.
- Investigated how Llama 3.1 prompting and repeated sampling affect issue resolution with oracle-retrieved context; experiments resolved 77 % of evaluated issues.
- Co-authored an arXiv preprint on test-feedback-driven program repair (2024).

Student Researcher — HSE Computational Biology Lab

Mar–Sep 2023

Worked in C++ on parallelization and concurrency optimizations for the Crux proteomics toolkit.

Technical Service

IOI 2025 Scientific Committee — Problem setting, testing, and contest design.

BSc Thesis — GFlowNet Post-training for Molecule Generation

Studied GFlowNet post-training of BioT5+ for molecule generation from text. Compared training on newly sampled token-generation sequences (trajectories) alone with adding supervised or replayed trajectories. Prioritized replay by molecular reward or trajectory-balance error, keeping the architecture and training objective fixed.

Honors and Competitions

- ICPC World Finalist (2022); Topcoder Open South America Regional — 2nd place (2019).
- ICPC South America Regional — 3rd (2020), 10th (2019), and 15th (2018); Bolivia Preliminary Contest — 1st place (2018, 2019, 2020, and 2025).
- International Olympiad in Informatics (IOI) — participant (2014, 2015, and 2017).
- Ibero-American Olympiad in Informatics — bronze (2014 and 2015); Bolivian Olympiad in Informatics — gold (2016) and silver (2013, 2014, and 2015).

Technical Skills

Programming Python · C++ · SQL · Bash

Software FastAPI · PostgreSQL · Docker · Git/GitHub · GitHub Actions

ML / Search PyTorch · Transformers · LoRA/QLoRA · FAISS

Vision OpenCV · YOLO · ByteTrack · ONNX Runtime

Languages: Spanish (native) · English (fluent) · Russian (intermediate)

Teaching and Coaching. Taught algorithms and data structures at UMSS (2025), introductory machine learning at UMSS (2026), and private online LLM classes (2025). Coaching activities in Bolivia (2016–2023) included school students and ICPC team co-coaching; taught algorithms, problem solving, and contest strategy at UMSS ICPC camps (2018–2019).